Alec Plehn

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EDUCATION

Brigham Young University, Ira A. Fulton College of Engineering Expected Apr 2026
Bachelor of Science, Mechanical Engineering, Emphasis in Aerospace Provo, Utah

LEADERSHIP & SERVICE

- Eagle Scout: Led logistics and assembly of 250 welcome kits for Syrian refugee families in Houston
- Provo Bicycle Collective: Performed drivetrain, wheel, and bearing service; trained volunteers

ENGINEERING PROJECTS

The Boeing Company, BYU Capstone Project Sep 2025 – Present
Team Secretary and Design Engineer Provo, Utah

- Collaborated with Boeing engineers on the design of a manufacturing system related to automation of wire harness production. Progressed the design to a TRL 4.
- Coordinated technical documentation, meeting records, and milestone tracking in Microsoft Teams.
- Lead rapid prototyping efforts to support concept validation and design iteration. This involved pneumatic system design and fabrication of components with CNC machining and 3D printing
- Modeled 15 custom prototype components and created assemblies in Solidworks. Performed fatigue analyses in ANSYS. Created part drawings for communication with stakeholders and patent attorneys.

BYU Mechatronics Projects

- Created an autonomous line-following, block-stacking robot. Implemented PID control on Arduino
- Designed and built a line-following robot capable of sorting colored balls into bins, navigating off-line, and performing other tasks within a 3-minute time window in collaboration with a team of 4.
- Implemented microcontroller-based PID control using both Arduino and PIC microcontrollers, fabricated components via 3D printing and PET thermoforming.

BYU Aeronautics Project

- Designed a controlled foam glider to achieve a par score in competitive "glider golf." Used python tools to optimize wingspan, airfoil selection, thickness, and chord distribution among other parameters.
- Fabricated components using CNC processing and additive manufacturing techniques.

RELEVANT EXPERIENCE

Northwest Services Assembly Feb 2022 – Present
Assembly / Pre-delivery Inspection Technician Salt Lake City, Utah

- Trained technicians across 5 store locations in standardized assembly and inspection practices
- Developed and implemented efficient product assembly processes while maintaining quality standards
- Performed mechanical repairs and coordinated replacement parts for retail hardware equipment
- Recognized by management at 4 Lowe's and Home Depot locations for reliability and workmanship. References available upon request

Huma Inc. Mar 2021 – Feb 2022
Bottling Line Technician Gilbert, Arizona

- Operated and maintained automated bottling and packaging equipment for continuous production.
- Supported plant improvement initiatives involving mechanical systems and production workflow.
- Recognized by company leadership for initiative and reliability.

TECHNICAL SKILLS

- **Proficient:** MATLAB, LabView, Python (Sympy, Scipy, Simulation), ANSYS, Solidworks, 3D printing, Rapid prototyping, Mechanical assembly, Arduino, Pneumatics
- **Familiar:** C++, R, PIC microcontrollers, Casting, Metal and plastics casting, CNC machining, Thermoforming, Control systems, Optimization
- **Relevant Coursework:** Control Systems, Optimization, Flight Vehicle Design, Finite Element Analysis, Plastics Materials and Processing